Optimizing Innovation Diffusion in the Catalan Research System: A Network-Based Approach

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1 Introduction

In the last few decades, developed economies and societies around the globe have experienced a profound transformation process away from the traditional industrial economic model, and towards a new paradigm where innovation and creativity play a central role (Ruelas-Gossi (2020); Florida (2005)). The innovation landscape has also changed dramatically, as what previously were isolated silos of innovation have turned into interconnected nodes in a proper innovation ecosystem. This shift can be attributed, in part, to the necessities of the new economic model, but also to a generalized trend that increasingly values open innovation (Katz and Wagner (2014)), and to the notion that innovation is rooted in place (Florida (2005); Barceló (2023)). In this context, regional innovation systems have emerged, and networks have been formed.

Governments are now tasked with the complex mission of leveraging innovation ecosystems to advance their industrial, economic, and social agenda, and ultimately, to maximize social welfare. In practical terms, governments might be interested in how to maximize the return on public investment in R&D, or in how to design effective policies that act as an innovation catalyst.

In this paper, we approach a particular challenge, namely, how to design a public policy that maximizes the spread of information in the Catalan regional innovation system. To do so, we use network theory, provided that we consider it to be a valuable framework from which to approach innovation ecosystems, considering that they are networks in essence. Specifically, we ask the following question: Which specific inter-institutional relationships within the Catalan research system should be strengthened in order to maximize the diffusion of innovative ideas within each domain-specific research network?

To answer this question, we construct four network layers: one for each major research domain. Our motivation for generating segregated networks comes from the fact that we deem of interest simulating how similarly or dissimilarly is knowledge transmitted across domains. Additionally, many research centers in the Catalan ecosystem are highly specialized in one field, which makes the analysis of the general network lose relevance for policy-making.

In these networks, nodes represent research institutions located in Catalonia, while edges represent the probability that a researcher in one institution knows at least one researcher in the other, which we interpret as a proxy for the probability of knowledge transmission. Notice that the focus of this paper is not on the nodes, but on the edges, that is, in the institutional relationships. Our interest in establishing the edges as the unit of interest comes from the fact that they have not been as extensively explored as nodes, but also from the fundamental notion of connectedness that innovation has come to hold.

1.1 Literature Review

The study of innovation diffusion through network theory is rooted in the seminal work of Granovetter (1973) on the "strength of weak ties," which posits that efficient pathways for information propagation often lie in non-redundant connections between otherwise disconnected clusters. In regional research systems, this perspective has evolved into the study of knowledge spillovers. Audretsch and Feldman (1996), demonstrate that spatial and institutional proximity significantly influence the rate at which knowledge flows across an economy.

Recent literature further emphasizes that knowledge diffusion is not uniform but domain-specific. The collaborative structure of Life Sciences, for instance, often differs substantially from that of Physical Sciences or Engineering (Borgatti and Halgin (2011)). Moreover, research on network topology suggests that the overall reachability of innovation is shaped by the presence of central hubs and by the probability of inter-institutional knowledge transfer (Balland, Boschma, and Frenken (2015)). Despite these advances, most studies still treat innovation primarily as a static output rather than as a dynamic diffusion process. Policy interventions frequently prioritize supporting top-performing institutions; however, evidence suggests that bridging structural holes between otherwise disconnected research centers may be more effective in facilitating system-wide knowledge propagation (Burt (2004)). As we introduced in the previous section, this is why we decide to focus our study in the edges of the networks and not in the nodes themselves.

1.2 Expected Contribution

This research makes three main contributions:

1. **Network construction and Domain-specific insights:** We construct several networks of the Catalan research ecosystem. By constructing separate network layers for different scientific domains, we capture how patterns of inter-institutional connectivity vary across fields. This allows us to identify, within each domain, the specific institutional relationships whose reinforcement would increase the diffusion potential of new ideas the most, and also to explore potential dissimilarities across knowledge domains.
2. **Methodological contribution:** We propose a probabilistic framework for modeling inter-institutional connectivity, moving beyond simple co-publication counts to estimate the likelihood of knowledge transmission between institutions.
3. **Policy implications:** We provide a data-driven prioritization framework for science policy. Rather than recommending broad funding strategies, we identify specific "bottleneck" edges whose reinforcement would generate the greatest marginal increase in the systemic reach of potential innovations.

2 Networks construction

2.1 Data

To the best of our knowledge, a domain-specific or system-wide network of Catalonia's research and innovation ecosystem is not publicly available. Thus, we generate our own networks using co-authorship in scholarly output to proxy the probability that researchers from any two given Catalan institutions collaborate, which we understand to constitute a channel of knowledge transmission. We use OpenAlex as an open-access catalog of works, which we filter to select only the relevant works for our implementation (OpenAlex (2026)).

First, we generate a list of all research institutions in Catalonia. To do so, we first restrict the OpenAlex API query to institutions located in Spain, in order to reduce the search space. Since there is no field in OpenAlex that directly identifies the subnational region of an institution, we use the latitude and longitude associated with each institution and check whether they fall within an approximate polygon representing the territorial boundaries of Catalonia.

Second, for each of the Catalan institutions identified, we obtain all the works published with at least one author affiliated with that institution. Additionally, we restrict the sample to works published since 2025, not only due to concerns about the size of the resulting network, but also because our goal is not to provide a representation of the dynamics of the Catalan innovation ecosystem, but rather to simulate how knowledge transmission takes place at a specific point in time.

Finally, we obtain a dataset of all works published since 2025 that include at least one co-author affiliated with a Catalan research institution. The dataset is structured at the paper level: for each work, we record its domain classification and construct a mapping of authors to their institutional affiliations. Specifically, each paper contains a list of its authors, and for each author, we store the (potentially multiple) institutions with which they are affiliated. This results in a structured representation in which each observation corresponds to a paper and includes both its domain and a dictionary linking each author to their set of affiliations.

2.2 Methodology

With the data obtained in the previous section, we construct four separate networks, one for each domain. The OpenAlex website divides all works into four domains: Physical Sciences, which includes all works in the branch of natural sciences that studies non-living systems (such as Chemistry, Mathematics, or Physics); Social Sciences, which is the branch of science focused on societies and the relationships among individuals within them (such as Business, Economics, or Psychology); Health Sciences, which includes all works focused on human health and disease prevention, diagnosis, treatment, and management (such as Medicine, Veterinary, or Dentistry); and finally Life Sciences, which is considered the branch of science that involves the study of living organisms, such as microorganisms, plants, and animals, including human beings (such as Biochemistry, Genetics, or Pharmaceutics).

First, we filter the dataset, retaining only the works that belong to the corresponding domain. We then define as nodes all the institutions that appear in at least one paper within the selected domain.

To compute the edges, we follow a two-step procedure. First, we construct an author-level co-authorship network. For each domain, we extract all authors appearing in the filtered dataset and build a directed adjacency matrix , where each entry $A_{ij} = 1$ if authors i and j have co-authored at least one paper, and $A_{ij} = 0$ otherwise. Therefore, we assume two researchers to have a connection (know each other) if they have collaborated. Notice that we assume that by collaborating more than once, two researchers do not have a stronger connection. Similarly, we also assume that by collaborating with many peers from another institution, as opposed to just one peer, the connection is not diminished.

Second, we link authors to institutions by constructing a bipartite affiliation matrix M, where rows represent institutions and columns represent authors. An entry $M_{ai} = 1$ indicates that author i is affiliated with institution a, allowing authors to be associated with multiple institutions.

Using these two matrices, we aggregate connections from the author level to the institution level. Specifically, for each pair of institutions (a, b), we compute the number of authors in institution a who have collaborated with at least one author in institution b. This is implemented by propagating co-authorship connections through the bipartite structure and converting them into a binary indicator of whether a connection exists.

We then define the directed edge weight p_{ab} as the probability that a randomly selected author from institution a has collaborated with at least one author from institution b. Formally:

$$p_{ab} = \frac{1}{n_a} \sum_{i \in a} I \left(\sum_{j \in b} A_{ij} > 0 \right) \tag{1}$$

It is important to note that the resulting networks are directed, as in general $p_{ab} \neq p_{ba}$. This asymmetry reflects differences in institution size and collaboration patterns, since the probability is normalized by the number of authors in the origin institution.

Finally, after constructing the full institution-level matrices, we restrict the analysis to Catalan institutions by filtering both rows and columns accordingly, obtaining a domain-specific directed weighted adjacency matrix representing the probability of knowledge transmission within the Catalan research network.

3 Modelling

To model the propagation of innovative ideas across the Catalan research system, we assume that knowledge diffusion follows an *Independent Cascade Model* (ICM), a standard probabilistic model used to study information spread in networks.

Let $G = (V, E)$ be a directed weighted network with the structure described in the previous section, where the weight of an edge from a to b, denoted by p_{ab}, represents the probability that a researcher from institution a has collaborated with (and thus knows) at least one researcher from institution b.

In the Independent Cascade Model, diffusion unfolds in discrete time steps. Initially, a small set of nodes is activated (the seed nodes), representing the institutions where an innovative idea originates. At each time step, newly activated nodes have a single opportunity to activate each of their inactive neighbors. The activation attempt succeeds with probability p_{ab}. If the attempt fails, the edge cannot be used again in subsequent steps. The process continues until no further activations occur.

In the context of our study, activation corresponds to an institution adopting or becoming aware of a new idea. The probabilistic transmission mechanism captures the uncertainty inherent in informal scientific interactions and knowledge spillovers between researchers.

3.1 Policy Simulation

Our objective is to identify which inter-institutional relationship should be strengthened in order to maximize the diffusion of ideas in the network. We model a potential policy intervention as an increase in the probability of interaction between two institutions.

Formally, for each unordered pair of vertices (a, b), we simulate a policy that increases the transmission probabilities in both directions by a constant value c. If an edge already exists, its weight is increased; otherwise, a new (virtual) edge is introduced. That is:

$$p'_{ab} = p_{ab} + c \quad \text{and} \quad p'_{ba} = p_{ba} + c$$

All other probabilities remain unchanged. In the simulations, newly created edges are treated equivalently to existing ones, with activation probabilities given by c.

For each modified network, we estimate the diffusion potential using Monte Carlo simulations of the Independent Cascade Model. Specifically, we use all nodes as seed nodes, and for each seed, we run two independent realizations of the diffusion process to account for stochastic variability. The outcome of each simulation is measured by the total number of institutions that become activated

during the diffusion process.

For each pair (a, b), we then compute the average diffusion reach across all simulations under the modified probabilities p'_{ab} and p'_{ba}. This provides an estimate of the expected influence of strengthening the relationship between institutions a and b. The pair that yields the largest increase in expected diffusion can therefore be interpreted as the most impactful target for policy intervention.

Note: Since we use all n nodes as seeds and perform two independent realizations for each seed, each edge is effectively evaluated through $n \times 2$ simulations, where n is the number of institutions in the domain-specific network. This ensures a robust estimation of the expected diffusion but increases computational demand.

3.2 Policy Definition

We understand the exact value of c to depend on the specific policy to be implemented. In order to introduce some rigor into the selection of this value, we first theoretically define which policy change will be taking place.

In our case, we define the policy to be implemented as the creation of a joint research program between institutions a and b. This new program will affect all potential pairs of institutions homogeneously, that is, that the effect of c will be independent of the institution size. Consequently, notice how the parameter c enters the equation defined in the section above as an addition, rather than through any other mathematical operand.

We set the effect of the newly created joint program as $c = 0.2$, which means that the probability that a researcher from a knows someone from institution b increases by 0.2. Similarly, we expect the effect to be the same the other way around. The value is chosen to represent a substantial but plausible increase.

4 Results

In this section, we present the networks resulting from the methodological construction process explained above, along with the results of the simulation exercise for each of the knowledge domains.

Note that the figures in this section, which represent each of the networks, have edge width proportional to the corresponding transmission probability. There are always two edges between any pair of nodes (one from each node to the other) since the graph is directed.

Additionally, also notice how in the tables where we display the simulation results, the average p and the node degree correspond to those of the out-going edges, that is, of the institutional relations that point outwards.

Physical Sciences Network

The Physical Sciences collaboration network, comprising 319 nodes and 4,664 edges, shows a structure characterized by a dominant connected component alongside a number of less integrated or isolated institutions. Despite the relatively broad base of participating institutions, the network is notably sparse. This is reflected by the lower density (0.09) and an average degree of 58.48, indicating that out of the tens of thousands of potential collaborative ties, only a small fraction is realized. This points to a highly selective pattern of connections among institutions, where collaborations may be concentrated around specific specialized facilities or projects.

Network Metric	Value
Number of Nodes	319
Number of Edges	4664
Density	0.09
Diameter	4
Clustering Coefficient	0.39
Average Path Length	2.02
Average Degree	58.48

Table 1: Summary of network metrics for the Physical Sciences domain.

Despite this lower connectivity and edge count relative to the network size, the structure still displays small-world characteristics. The diameter remains low at 4, and the average path length of 2.02 suggests that institutions can still reach each other through relatively few intermediaries, although slightly less efficiently than in other cases. The clustering coefficient of 0.39 points to a moderate level of local cohesion, indicating the presence of collaborative subgroups, although these are presumably less tightly interconnected than in the broader research ecosystem.

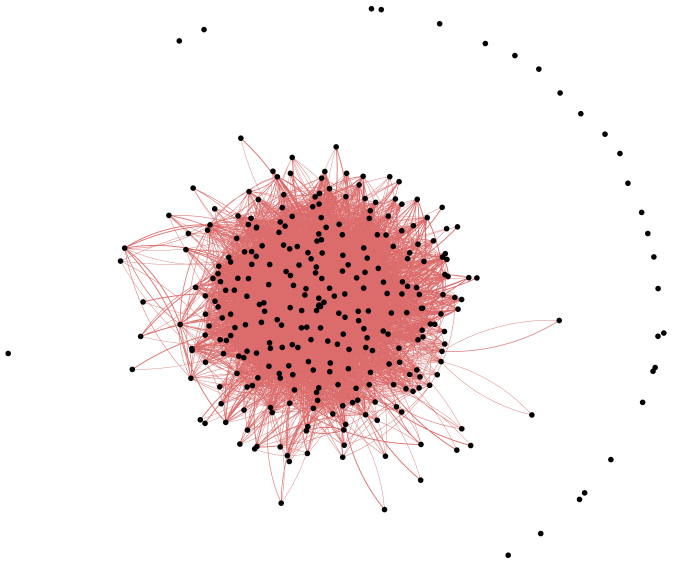


Figure 1: Physical Sciences Network

The policy simulation scenario specified in previous sections returns the pair of Catalan research institutions where a joint research program should be established, so that the diffusion of new ideas is maximized. The obtained results are:

Table 2: Top Institution Pairs in Physical Sciences Network by Average Diffusion

Rank	Institution A	Avg P	Degree	Institution B	Avg P	Degree	Distance	Avg Diff
1	Ficosa International (Spain)	0.58	2	S21sec (Spain)	0.00	0	∞	127.71
2	Generalitat de Catalunya	0.06	96	Ecosystem (Spain)	0.11	14	1	127.35
3	Elisava Barcelona School of Design and Engineering	0.09	8	Barcelona Provincial Council	0.23	16	2	127.33
4	Barcelona Supercomputing Center	0.02	84	Actar (Spain)	1.00	1	2	127.32
5	Spanish Multiple Sclerosis Network	0.50	6	Ultrasion (Spain)	1.00	5	2	127.32

It is worth noting that the differences in average diffusion across the top-ranked pairs are very small. This suggests that the policy recommendations may not be fully robust, as minor variations in the simulation outcomes or model parameters could lead to different pairs being identified as optimal.

For the physical sciences domain, the research centers to be targeted are the Ficosa International, and the S21sec. For these nodes, the mean average diffusion reached is of 127.71 nodes out of the total 319 institutions in this domain. At first glance, this result is somewhat counterintuitive, as S21sec has degree 0 and is therefore completely disconnected from the network.

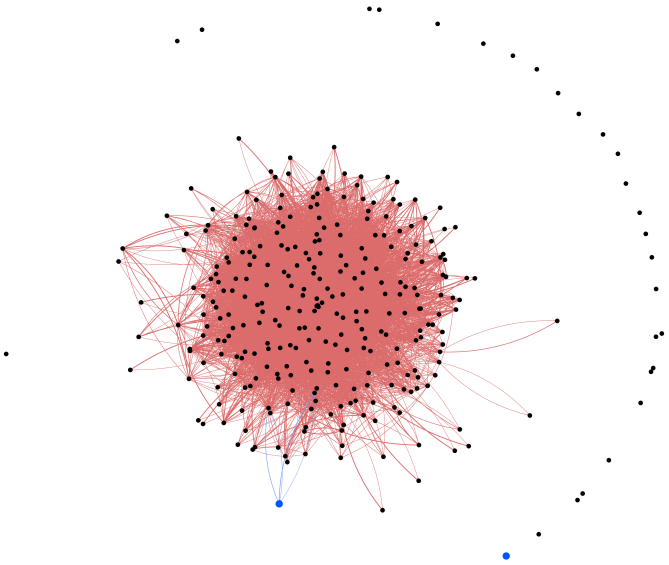


Figure 2: Physical Sciences Network, with the institutions to target highlighted in blue

This outcome can be understood in light of the overall diffusion dynamics observed in this network. In general, diffusion values are consistently high across the top-ranked pairs, suggesting that the system tends to amplify spreading processes regardless of the specific structural position of the nodes. As a result, in some cases it becomes optimal to establish links with otherwise isolated or external institutions. This can be explained by the fact that if an idea originates within the main connected component, it is very likely to spread throughout that component, whereas if it arises in an isolated node, it would otherwise remain confined and fail to diffuse.

More broadly, the results reveal a trade-off between transmission probability and network connectivity. On the one hand, institutions with very high transmission probabilities can generate strong diffusion even with very few links, since each connection is highly effective. On the other hand, institutions with lower transmission probabilities can still achieve comparable diffusion outcomes if they are highly connected, as their large number of links compensates for the lower efficiency of each individual interaction. This is illustrated in the table by the presence of both highly connected nodes (e.g., Generalitat de Catalunya, or Barcelona Supercomputing Center) and low-degree nodes with high transmission probabilities (e.g., Actar, or Ultrasion) among the top-ranked pairs.

Overall, these findings suggest that, in this network, it may not always be optimal to apply the policy between already well-connected institutions. Instead, greater gains in diffusion may be achieved by strategically linking central nodes with peripheral or isolated ones, thereby expanding the reach of ideas beyond the main connected component. This highlights the importance of considering not only the existing structure of the network but also the potential benefits of integrating

otherwise disconnected actors.

Health Sciences Network

The Health Sciences collaboration network is the most highly interconnected of the domains analyzed, comprising 324 nodes and 11,890 edges. With a dominant giant component and very few disconnected institutions, the volume of collaborative ties is substantial. Its density (0.23) and remarkably high average degree (146.79) indicate a rich pattern of collaborations. This suggests that the 324 institutions in this domain are extensively linked, likely through well-established partnerships, large-scale clinical trials, and central hubs that facilitate widespread coordination and knowledge exchange.

Network Metric	**Value**
Number of Nodes	324
Number of Edges	11890
Density	0.23
Diameter	4
Clustering Coefficient	0.59
Average Path Length	1.78
Average Degree	146.79

Table 3: Summary of network metrics for the Health Sciences domain.

The network also exhibits strong small-world properties. With a diameter of 4 and a very low average path length of 1.78, institutions are separated by only a few steps, enabling highly efficient diffusion of information and practices. The clustering coefficient (0.59) is relatively high, pointing to tightly knit collaborative groups and a strong tendency toward local cohesion. Overall, the structure reflects both robust global connectivity across the entire node set and dense local clustering within the domain.

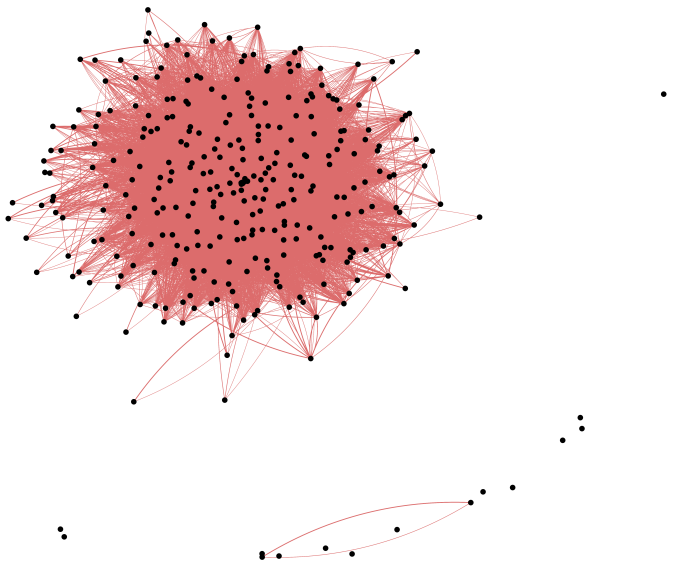


Figure 3: Health Sciences Network

The policy simulation scenario specified in previous sections returns the pair of Catalan research institutions where a joint research program should be established, so that the diffusion of

new ideas is maximized in the Health Sciences domain. The results obtained are displayed in the following table.

Table 4: Top Institution Pairs in Health Sciences Network by Average Diffusion

Rank	Institution A	Avg P	Degree	Institution B	Avg P	Degree	Distance	Avg Diff
1	Universitat Ramon Llull	0.04	131	Neuraxpharm (Spain)	0.33	3	2	184.13
2	Hospital Plató	0.12	39	Boehringer Ingelheim (Spain)	0.05	34	2	184.08
3	Center for Research in Environmental Epidemiology	0.13	69	Hydrometeorological Innovative Solutions (Spain)	0.00	0	∞	183.83
4	Otsuka Pharmaceutical (Spain)	0.01	11	Ajuntament de Barcelona	0.27	12	2	183.80
5	Agència de Qualitat i Avaluació Sanitàries de Catalunya	0.10	99	Medtronic (Spain)	0.33	1	2	183.72

For the health sciences domain, the results indicate that the policy should be applied at the relationship between Universitat Ramon Llull and Neuraxpharm. For these nodes, the mean average diffusion reached is of 184.13 nodes out of the total 324 institutions in this domain.

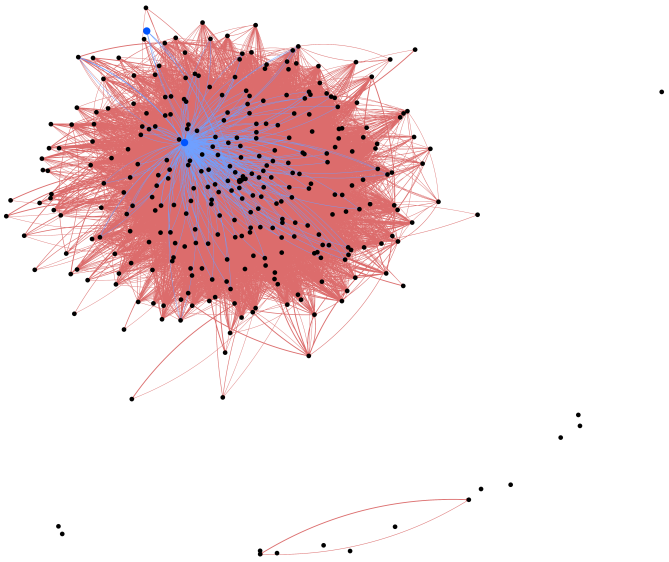


Figure 4: Health Sciences Network, with the institutions to target highlighted in blue

The simulation results reveal a consistent approach for maximizing knowledge diffusion in the Health Sciences domain. Rather than recommending new connections between established, high-degree academic hubs, the model heavily prioritizes cross-sectoral bridges between central public/academic institutions and specialized industry players.

This dynamic is perfectly illustrated by the top-ranked intervention. The simulation pairs Universitat Ramon Llull, a massive academic hub (degree 131), with Neuraxpharm, a private pharmaceutical company. Notably, while Neuraxpharm operates on the periphery with a very low degree (3), it possesses a remarkably high average outward transmission probability (0.33). By forging this link, the policy effectively leverages the private firm as a high-efficiency "gateway," allowing innovations generated at the central university to rapidly cascade into specific commercial and clinical clusters.

This academia-to-industry pattern holds true across the top five interventions, which consistently pair public research centers, agencies, or hospitals (such as the Hospital Plató, or the Agència de Qualitat i Avaluació Sanitàries de Catalunya) with private medical, technological, or pharmaceutical firms (such as Boehringer Ingelheim, or Medtronic).

Furthermore, the mathematical logic of the Independent Cascade Model highlights the immense value of inclusive innovation. This is particularly evident in the third-ranked intervention, which targets Hydrometeorological Innovative Solutions, an isolated node. By connecting a well-integrated public research center (Center for Research in Environmental Epidemiology) to a completely disconnected industry actor, the simulation demonstrates that unlocking peripheral components yields some of the highest marginal gains in overall network reach.

Ultimately, these results suggest that in an already highly cohesive ecosystem like Health Sciences, policy resources are best allocated toward cross-sectoral integration rather than reinforcing the existing academic core. Bridging the gap between high-volume public research hubs and highly efficient, specialized industry nodes allows new ideas to spread in the network, successfully reaching over 56% of all institutions in the domain.

Life Sciences Network

The Life Sciences collaboration network presents a well-connected structure consisting of 292 nodes and 7,534 edges, likely dominated by a central giant component alongside a smaller number of less integrated institutions. Generating nearly eight thousand collaborative ties among its actors, the network exhibits a substantial level of collaboration, though not as extensive as in the Health Sciences domain. With a density of 0.18 and an average degree of 103.21, this suggests a balanced structure where institutions maintain a considerable number of ties without reaching the highest levels of structural saturation.

Network Metric	**Value**
Number of Nodes	292
Number of Edges	7534
Density	0.18
Diameter	3
Clustering Coefficient	0.54
Average Path Length	1.86
Average Degree	103.21

Table 5: Summary of network metrics for the Life Sciences domain.

The network retains clear small-world characteristics, with a diameter of 3, the smallest among all domains, and an average path length of 1.86, indicating that the 292 institutions remain relatively close to one another in terms of collaborative steps. The clustering coefficient of 0.54 points to a fairly strong tendency toward local cohesion, with institutions forming interconnected groups, although these are somewhat less tightly clustered than in the most cohesive domains. Overall, the network combines moderate-to-high connectivity with solid local clustering, facilitating steady scientific collaboration.

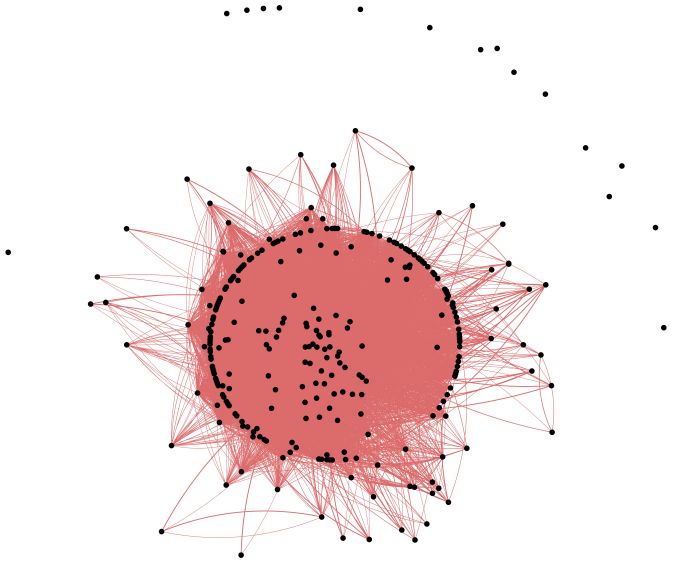


Figure 5: Life Sciences Network

For the Life Sciences domain, we again run a simulation to find the pair of institutions where a joint research program should be established so that the diffusion of new ideas is maximized. The results obtained are displayed in the following table.

Table 6: Top Institution Pairs in Life Sciences Network by Average Diffusion

Rank	Institution A	Avg P	Degree	Institution B	Avg P	Degree	Distance	Avg Diff
1	IESE Business School	0.00	0	Museu de Ciències Naturals de Barcelona	0.08	52	∞	166.75
2	Barcelona Institute of Science and Technology	0.08	85	Starlab Barcelona SLU (Spain)	0.19	8	2	166.34
3	Oryzon Genomics (Spain)	0.18	69	APC Europe (Spain)	1.00	4	3	166.32
4	EAE Business School	0.12	26	Instituto de Microcirugía Ocular	0.20	5	2	166.03
5	Hospital Sant Joan de Déu Barcelona	0.05	137	Barcelona Macula Foundation	0.13	33	1	166.01

In this case, the simulation shows that the inter-institutional relationship to be increased is the one between IESE Business School and Museu de Ciències Naturals de Barcelona. Strengthening the relationship between these two centers, we obtain a mean average diffusion of 166.75 nodes out of the total 292 institutions in this domain.

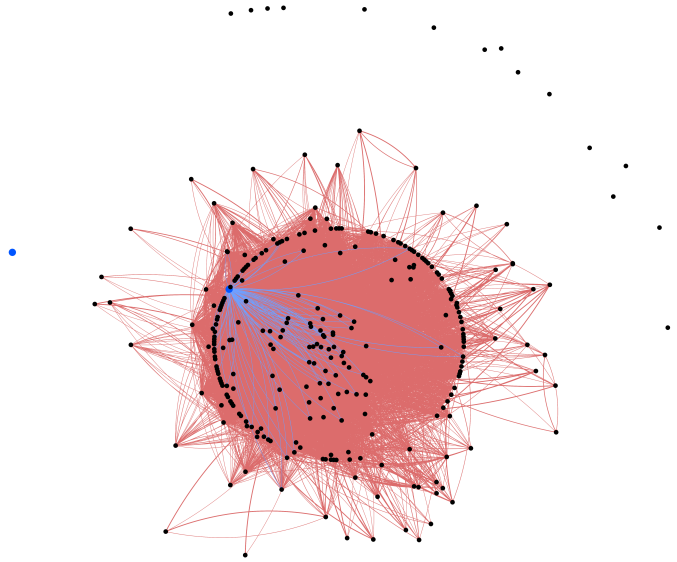


Figure 6: Life Sciences Network, with the institutions to target highlighted in blue

As in the Physical Sciences case, this result may initially appear counterintuitive, as IESE Business School has degree 0 and is therefore disconnected from the main network. However, this result is closely linked to the structural properties of the Life Sciences network. Due to its high connectivity, low diameter, and relatively strong clustering, diffusion within the main component is already highly efficient. This means that most institutions are easily reachable from one another, and ideas can spread rapidly without the need for additional links within the core of the network. As a consequence, reinforcing existing connections tends to generate diminishing returns.

In contrast, isolated or weakly connected institutions represent untapped potential. Establishing a link between a disconnected node and a well-integrated institution effectively allows ideas originating outside the main network to enter a system where diffusion is already highly efficient. In this sense, the policy is not so much about improving internal transmission, but about extending the boundaries of the network.

At the same time, the results also reflect different mechanisms through which diffusion can be achieved. Some high-ranking pairs combine highly connected institutions, leveraging their central position, while others involve nodes with relatively few links but higher transmission probabilities, making each connection more impactful. This indicates that diffusion performance depends on how these two dimensions interact rather than on a single dominant factor.

Overall, in a network with such strong small-world properties, the most effective strategy is not necessarily to densify the already cohesive core, but to bridge structural gaps. By incorporating peripheral or disconnected institutions into the main component, the system can achieve broader and more inclusive diffusion outcomes.

Social Sciences Network

The Social Sciences collaboration network appears relatively sparse, representing the smallest domain in our study with 285 nodes and only 4,146 edges. The network features a central connected component accompanied by a notable number of less integrated or isolated institutions. The low density (0.10) and average degree (58.19) indicate that these 285 institutions have formed far fewer realized collaborations compared to the other domains, suggesting a more fragmented and highly

selective pattern of connections. This may reflect the more individualized nature of research in certain social science disciplines.

Network Metric	Value
Number of Nodes	285
Number of Edges	4146
Density	0.10
Diameter	5
Clustering Coefficient	0.47
Average Path Length	2.04
Average Degree	58.19

Table 7: Summary of network metrics for the Social Sciences domain.

In structural terms, the comparatively low edge count results in a network that is slightly more dispersed. The diameter increases to 5 and the average path length (2.04) is higher, indicating that institutions are, on average, farther apart and that knowledge diffusion across the network may be somewhat less efficient. The clustering coefficient (0.47) reflects a moderate level of local cohesion, with some tendency to form collaborative groups, though these are less tightly interconnected than in more cohesive domains. Overall, the network combines lower connectivity with moderate clustering and slightly reduced structural efficiency.

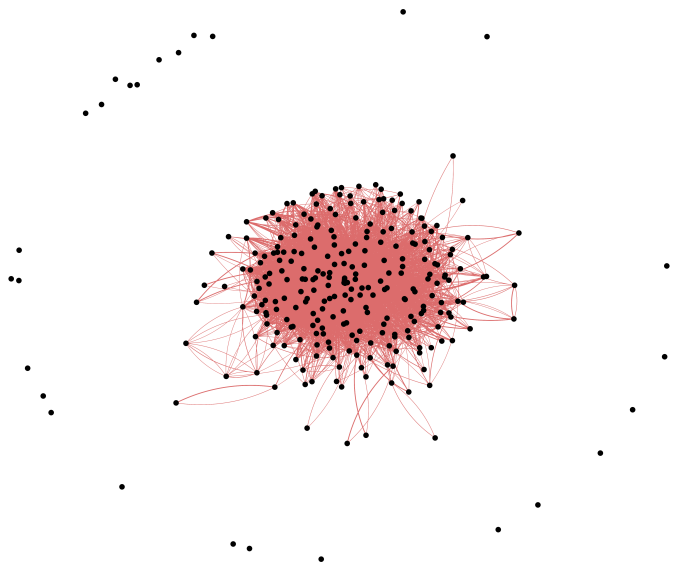


Figure 7: Social Sciences Network

The policy simulation scenario specified in previous sections returns the pair of institutions where a joint research program should be established, so that the diffusion of new ideas is maximized in the Social Sciences domain. The results obtained are displayed in the following table.

Table 8: Top Institution Pairs in Social Sciences Network by Average Diffusion

Rank	Institution A	Avg P	Degree	Institution B	Avg P	Degree	Distance	Avg Diff
1	Instituto de Investigaciones Químicas y Ambientales de Barcelona	1.00	9	Barraquer Opthalmology Center	0.15	23	2	113.82
2	Duran i Reynals Hospital	0.17	50	Catalonian Research and Innovation Centre	0.21	8	2	112.76
3	Centro de Investigación Biomédica en Red Diabetes y Enfermedades Metabólicas Asociadas	0.22	24	Quside Technologies (Spain)	0.35	17	2	112.59
4	IESE Business School	0.02	8	Catalonian Research and Innovation Centre	0.21	8	2	112.43
5	Institut Català de Paleontologia Miquel Crusafont	0.13	19	United Nations University Institute on Globalization, Culture and Mobility	0.08	1	3	112.30

For the social sciences domain, the research centers to be targeted are the Instituto de Investigaciones Químicas, and the Barraquer Opthalmology Center. For these nodes, the mean average diffusion reached is of 113.82 nodes out of the total 285 institutions in this domain.

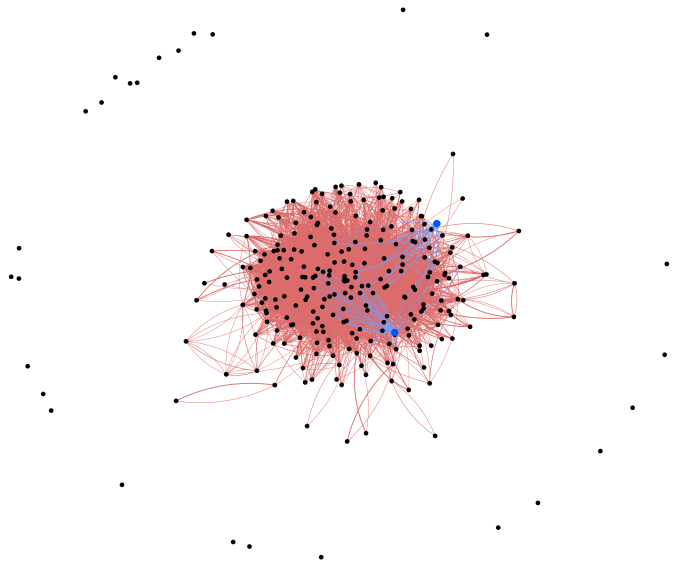


Figure 8: Social Sciences Network, with the institutions to target highlighted in blue

The results of the simulation highlight a clear strategy for maximizing diffusion in this relatively sparse and fragmented network. In the Social Sciences domain, the most effective policy interventions do not simply target the largest, most connected universities. Instead, the model prioritizes linking specialized research centers that act as highly efficient transmitters to moderately connected hubs.

This is evident in the top-ranked pair. The policy connects the Instituto de Investigaciones Químicas y Ambientales, a node with a low degree (9) but a perfect outward transmission probability (1.00), to the Barraquer Ophthalmology Center (degree 23). By forming this link, the policy uses the highly efficient chemical institute to push ideas reliably into Barraquer's broader network.

This pattern of cross-disciplinary bridging repeats across the top five results. Because the Social Sciences network is naturally more disconnected (with a diameter of 5), the best way to spread information is to connect distinct subgroups. For example, the Catalonian Research and

Innovation Centre (CRIC) appears twice in the top five, acting as a valuable bridge when paired with both a medical research facility (Duran i Reynals Hospital) and a business school (IESE).

Ultimately, the simulation shows that in a low-density domain, strategic bridges between different specialties are the most valuable investments. By establishing just one optimal connection between these distinct clusters, a policy intervention allows an innovation to reach roughly 40% of the entire network (113 out of 285 institutions), successfully overcoming the natural fragmentation of the Social Sciences domain.

5 Conclusions and Limitations

This study set out to evaluate whether targeted policy interventions (specifically, the creation or strengthening of links between selected pairs of institutions) can effectively enhance the diffusion of knowledge across different scientific domains. The results show that the answer is nuanced and highly dependent on the underlying structure of each network.

Across all domains, a consistent pattern emerges: diffusion levels are generally high, particularly in more cohesive networks such as Health Sciences and Life Sciences. In these cases, the presence of a well-connected giant component, short path lengths, and strong clustering already ensure that ideas can spread efficiently throughout most of the system. As a result, reinforcing existing links between well-integrated institutions tends to generate only marginal improvements in overall diffusion. In such contexts, the policy intervention may not be strictly necessary, as the network is already close to its maximum diffusion potential.

The simulations frequently identify optimal pairs that include isolated or weakly connected nodes. This reflects a key mechanism of the model: connecting a previously disconnected institution allows ideas originating there to access the entire network. However, from a policy perspective, these cases should be interpreted with caution. Although the relative gain in diffusion may appear significant within the model, the practical impact can be limited. If the isolated node has low activity, low transmission probability, or limited relevance within the research ecosystem, the real-world benefits of integrating it may not justify the cost of intervention.

In contrast, in more fragmented networks such as Social Sciences (and to a lesser extent Physical Sciences) policy interventions appear more justified. In these cases, the network structure exhibits lower density, higher diameter, and more pronounced fragmentation. Here, diffusion is less efficient, and the creation of strategic links between distinct subgroups or complementary institutions can significantly improve the reach of new ideas. Particularly effective are connections that bridge structural gaps, either by linking different clusters or by combining high-transmission nodes with moderately connected hubs.

Another key insight is the trade-off between transmission probability and connectivity. High-degree institutions offer broad reach but may have lower transmission efficiency, while low-degree institutions can still play an important role if their connections are highly effective. Optimal policy design therefore requires balancing these two dimensions rather than focusing exclusively on central or peripheral actors.

Overall, the findings suggest that a one-size-fits-all policy is not appropriate. In highly cohesive networks, resources may be better allocated toward maintaining existing structures or supporting the quality of interactions rather than creating new links. In contrast, in sparser or more fragmented domains, targeted interventions that bridge disconnected components or connect complementary actors can yield substantial gains in diffusion. Additionally, the very small differences in diffusion observed across the top-ranked pairs suggest that the resulting policy recommendations should be interpreted with caution. In practice, multiple alternative interventions may yield very similar outcomes, meaning that the choice of policy may not be uniquely determined by the model.

In conclusion, while the proposed policy can be effective, its implementation should be selective. Interventions are most valuable when they address genuine structural gaps in the network, and less so when they simply reinforce already efficient systems or connect marginal actors with limited practical impact.

Despite the methodological rigor of our approach, there are several limitations that should be acknowledged:

First, our analysis only includes papers published from 2025 onward. While this restriction helps manage the size of the dataset and focuses on recent diffusion knowledge, it omits historical collaborations that could have influenced the current structure of the Catalan research network. Consequently, our networks may not fully capture long-term relationships and cumulative knowledge flows between institutions.

Secondly, the categorization of works into four broad domains (Physical Sciences, Life Sciences, Health Sciences, and Social Sciences) may oversimplify the true diversity of research areas. Some disciplines with distinct collaboration patterns may be grouped together, potentially masking domain-specific interactions and limiting the ability to detect finer-grained diffusion dynamics.

Finally, in the Independent Cascade Model simulations, we run only two independent realizations per seed node due to computational constraints. This relatively low number of realizations may increase the variance of the estimated diffusion outcomes, meaning that the calculated influence of specific institutional pairs could be sensitive to stochastic fluctuations in the simulation process.

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